Assignment: 01 PST 31128

ChatBot Summary

Name: M.H.A.Dilsara Madhumal
Reg No: 19APP3882
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ChatBot Summary

Title: Multi-Response Chatbot

Introduction: The Multi-Response Chatbot is an interactive program designed to engage in conversations with users. Developed using Python, the chatbot employs regular expressions and predefined responses to generate appropriate replies based on user input.

Features:

Message Analysis: The chatbot analyzes user messages to determine the likelihood of predefined responses. It assesses the presence of recognized words and checks for the existence of required words to gauge the relevance of responses.

Response Generation: Based on the analysis, the chatbot selects the most probable response from a set of predefined options. It considers single responses and ensures the inclusion of required words for accuracy.

Flexible Expansion: The chatbot's response capabilities can be expanded by adding new responses and adjusting the recognition criteria. This flexibility allows for customization and adaptation to different conversational contexts.

Natural Language Processing: While the chatbot primarily relies on keyword recognition, it aims to provide a natural conversational experience by understanding and responding to common greetings, inquiries about well-being, and expressions of gratitude.

Functionality:

Greeting: The chatbot greets users with a friendly message when appropriate keywords such as "hello," "hi," or "hey" are detected.

Well-being Inquiry: It responds to inquiries about its well-being, demonstrating a conversational element that simulates human interaction.

Interest Recognition: By detecting keywords related to coding, the chatbot expresses appreciation when users mention their love for coding or specific platforms.

Eating Habit Inquiry: The chatbot answers questions about its eating habits, providing a predefined response to inquiries related to food consumption.

Implementation: The chatbot is implemented using Python, leveraging regular expressions for message parsing and predefined responses stored in external modules. It employs a modular approach, allowing for easy modification and expansion of response categories.

Conclusion: The Multi-Response Chatbot offers a simple yet effective solution for engaging in basic conversational interactions. With its ability to analyze messages, generate appropriate responses, and simulate natural language processing, the chatbot provides users with an interactive and dynamic experience. As an adaptable tool, it holds potential for further enhancements and integration into various applications requiring conversational interfaces.